

Claim Denial Prediction

Problem Framing

The review team can only manually inspect 25% of claims by risk core before submission. That capacity limit is what should drive the metrics and threshold choice in this project.

Given the constraint, the 2 possible errors are asymmetric in cost:

- False negative – a claim that will be denied but scores too low to make the reviewed 25%.
- False positive – a claim that is flagged as risky.

A missed denial is far more expensive than a waster review, the model was optimized to maximize recall with the fixed top ~ 25% budget rather than treating both error type as equally costly, the way a default 0.5 threshold or plain accuracy would

Data Findings

About 1 in 5 claims in denied, ie overall 21.6% in the training data.

Claims in which prior authorization was required but is missing from the file or where the required referral is missing are denied far more than the average. This is a strong driver for the denial risk in the data.

A model with zero training – ie a random classifier would catch about 25% of denials (randomly selecting 25% claims)

A trained model would capture around 46%

With trained model also at the top-25% cut off, there are chances that some of the predicted denials turn out to be a non-denial claim.

Model Built

A logistic regression based binary classifier is built. Used class_weight = balanced)

Some derived features like prior_auth_vs_has_auth – Whether the claim requires authorization and has a prior authorization.

& ref_req_vs_ref_present - Whethther referral is required and referral is present in the claim.

& payment_ratio – expected_payment upon total_billed
are used to add more dimensionality and adding more influence in the prediction technique.

Below are the some of the metrics reported.

At top refers to the 25% selection based on percenti

```
{
  "model": "logistic_regression",
  "seed": 42,
  "validation": {
    "pr_auc": 0.5094694098447391,
    "roc_auc": 0.6953813104189044,
    "top25pct_threshold": 0.5954353143720383,
    "top25pct_denials_predicted": 135,
    "top25pct_denial_percentage_of_total": 0.2504638218923933,
    "top25pct_precision_at_top": 0.4888888888888889,
    "top25pct_recall_at_top": 0.4714285714285714,
    "top25pct_denial_capture_rate": 0.4714285714285714
  },
  "test": {
    "pr_auc": 0.454059302137873,
    "roc_auc": 0.706488821590808,
    "top25pct_threshold": 0.5917324616381743,
    "top25pct_denials_predicted": 135,
    "top25pct_denial_percentage_of_total": 0.2504638218923933,
    "top25pct_precision_at_top": 0.37037037037037035,
    "top25pct_recall_at_top": 0.4854368932038835,
    "top25pct_denial_capture_rate": 0.4854368932038835
  }
}
```

